(a) Step 2a: Design Brief Selection

Propose & Select Brief (≤ 2 rev.)
Input: Stage 1 Skills (4.MD.A.3, 5.NF.B.4)

Student-Facing Setup (To Stage 3)

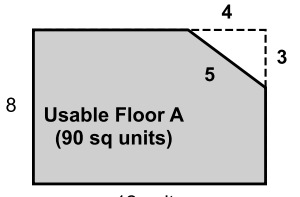
- Goal: Compare two play-area floor plans
- Rules: Identical outer boxes; A cuts a slanted corner, B cuts a rectangle corner
- Claim: "Same outer box = same area"

Internal Design Notes (Withheld)

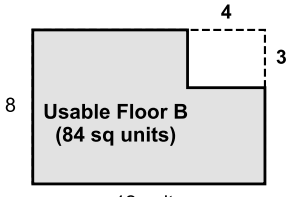
- Impasse: Outer-box areas tie (l × w)
- Route 1: Corner triangle = ½ of 4 × 3 box
- Route 2: Split floor into rectangles

(b) Selected Unsplit Geometry (6.G.A.1)

Design A: Slanted Corner Cut



Design B: Rectangular Cut



Shortcut Impasse: Both floors sit inside 12 × 8 = 96 outer rectangles, so comparing outer dimensions ties (96 = 96) unless students find a way to account for the slanted vs. rectangular corner cuts.

(c) Step 2b: Calculator Verification

Choose Numbers (≤ 2 repairs) ⇄ **Python Math Calculator**

Checked Conditions on Values (12, 8, 4, 3, 5)

1. Shortcut Fails (Outer Tie):
   $12 \times 8 = 96$ vs. $12 \times 8 = 96$ **[Holds]**
2. Consistent Right-Triangle Cut:
   $\sqrt{(4^2 + 3^2)} = 5$ **[Holds]**
3. Distinct Usable Floor Areas:
   $96 - 6 = 90 > 96 - 12 = 84$ **[Holds]**

Verified Given Values (12, 8, 4, 3, 5) → Stage 3